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(54) **EXTENDED UTILIZATION AREA FOR A MEMORY DEVICE**

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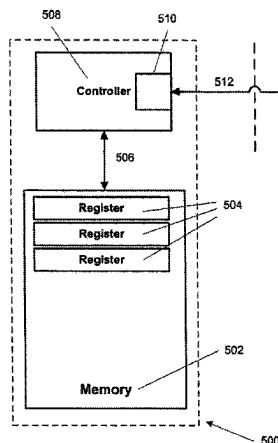
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ABSTRACT

Methods, systems and devices for configuring access to a memory device are disclosed. The configuration of the memory device may be carried out by creating a plurality of access profiles that are adapted to optimize access to the memory device in accordance with a type of access. For example, when an application with specific memory access needs is initiated, the memory access profile that is designed for that particular access need may be utilized to configure access to the memory device. The configuration may apply to a portion of the memory device, a partition of the memory device, a single access location on the memory device, or any combination thereof.

20 Claims, 5 Drawing Sheets

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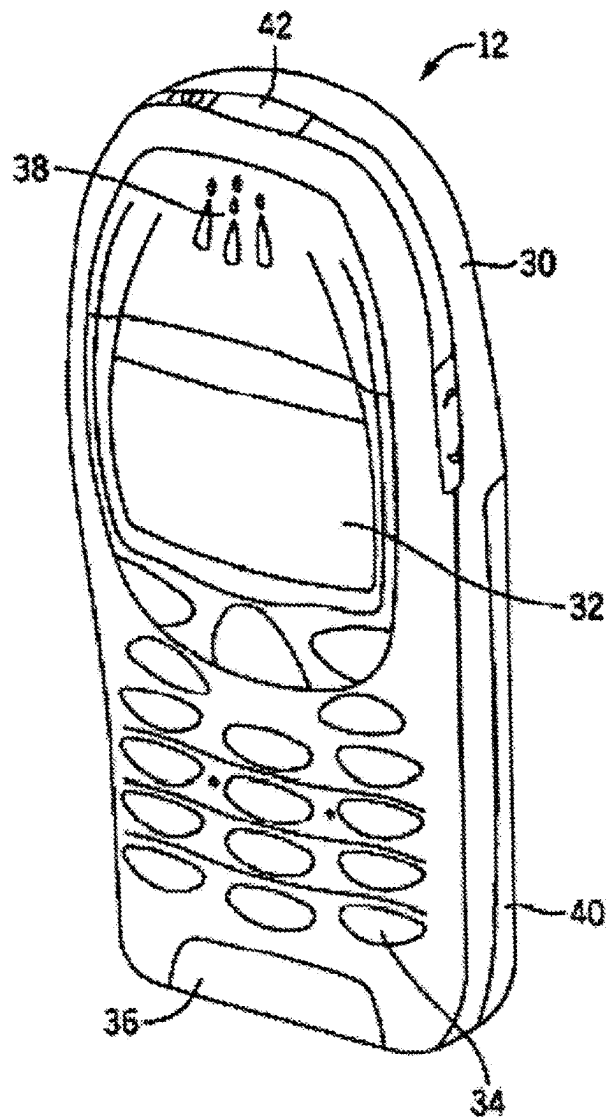


FIG. 1

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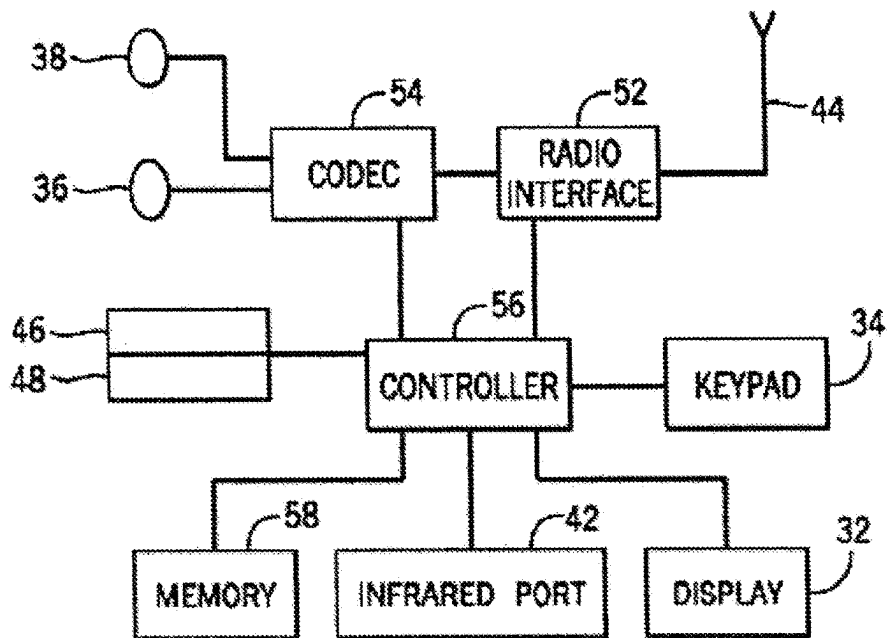


FIG. 2

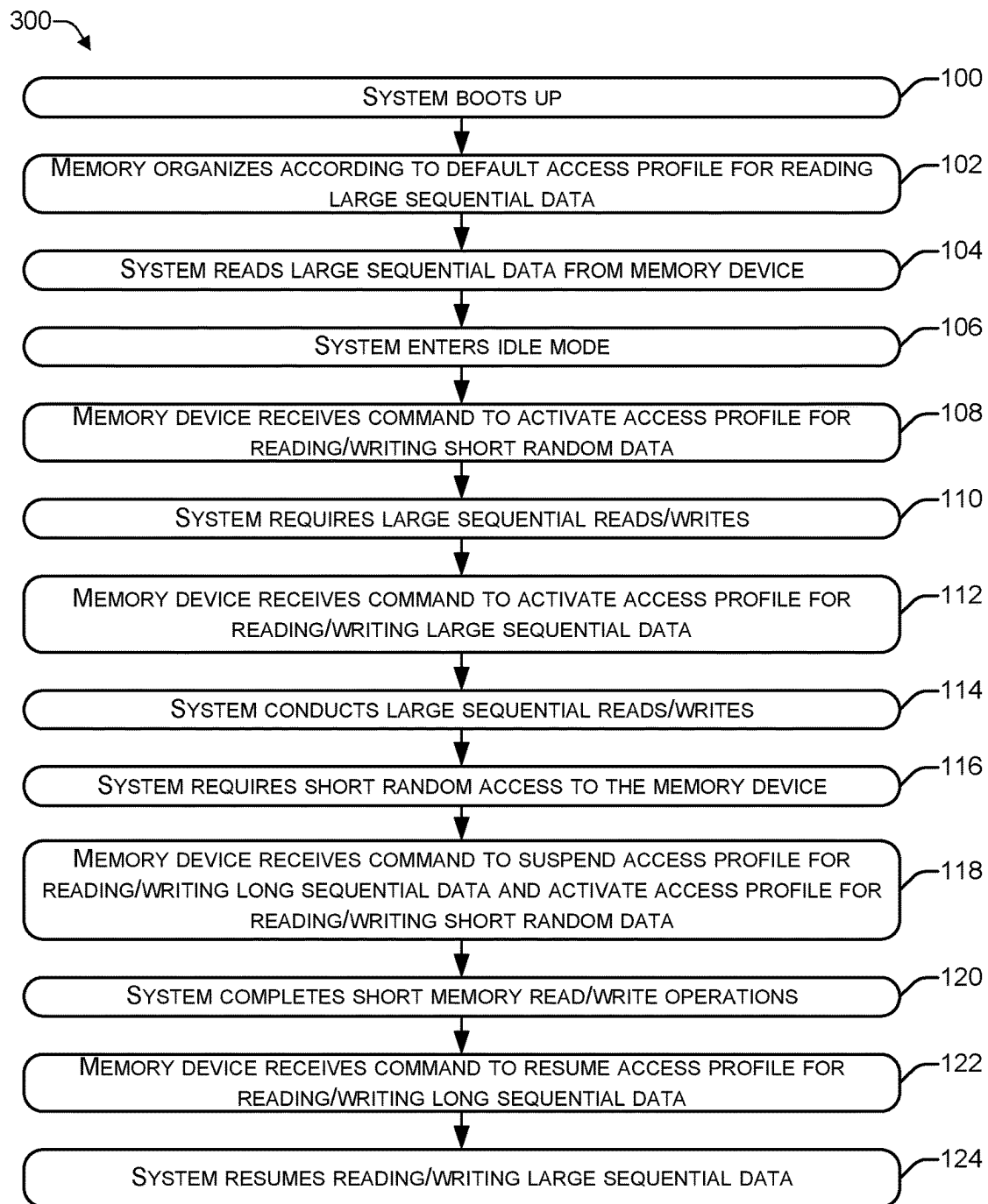


FIG. 3

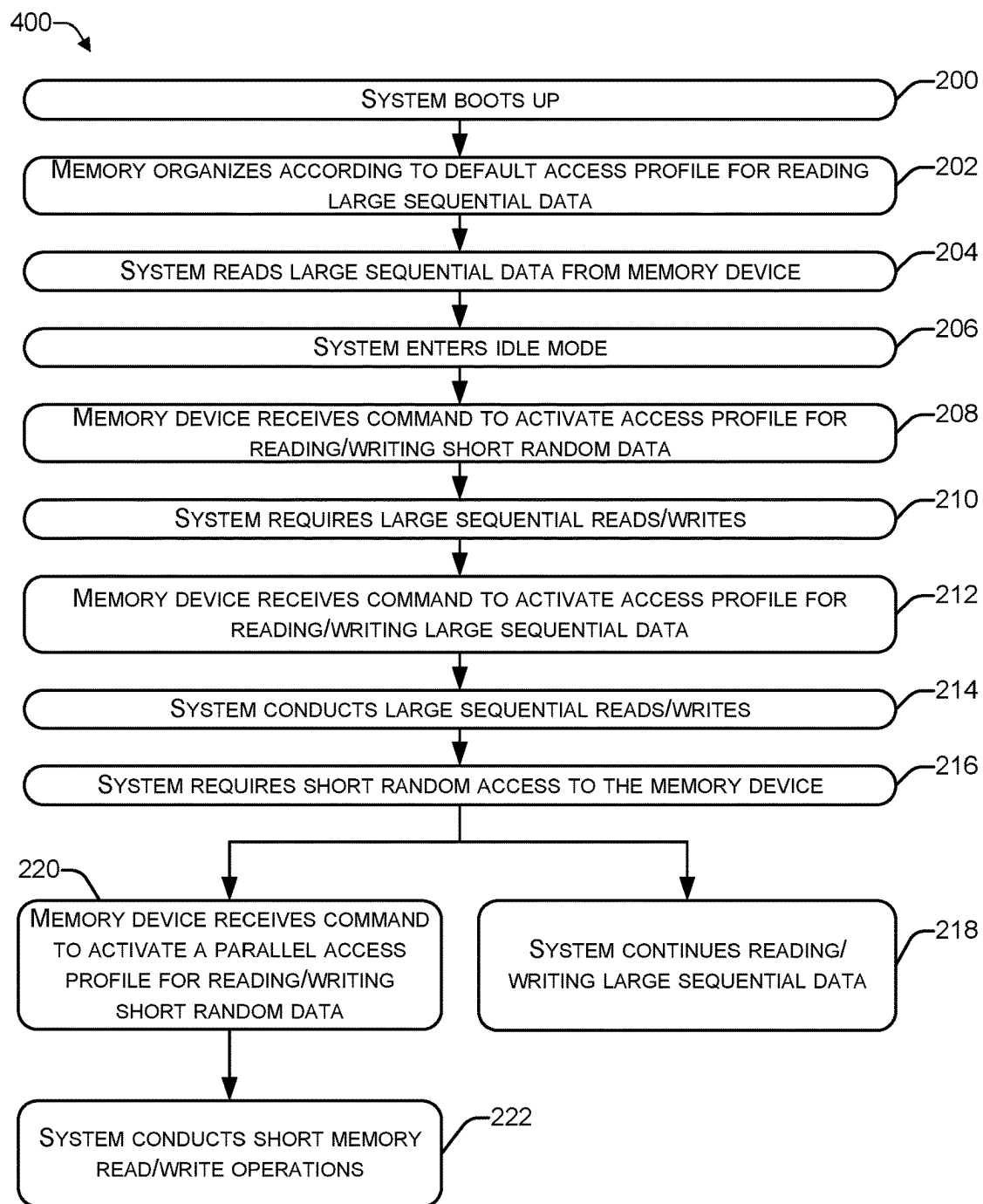


FIG. 4

500

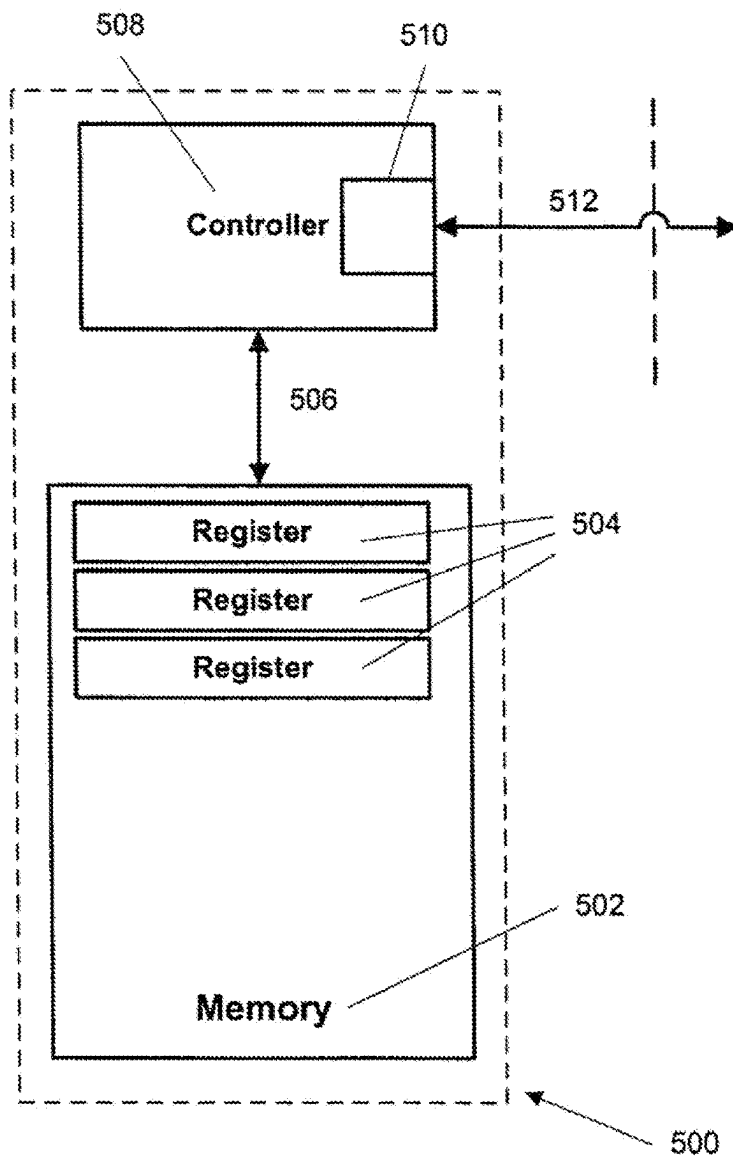


FIG. 5

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EXTENDED UTILIZATION AREA FOR A MEMORY DEVICE

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of and claims priority to pending U.S. patent application Ser. No. 18/151,977, filed Jan. 9, 2023, which is a continuation of U.S. patent application Ser. No. 18/045,067, filed Oct. 7, 2022, which is a continuation of U.S. patent application Ser. No. 17/410,588, filed Aug. 24, 2021, which is a continuation of U.S. patent application Ser. No. 16/825,653, filed Mar. 20, 2020 and issued as U.S. Pat. No. 11,182,079, which is a continuation of U.S. patent application Ser. No. 16/598,894, filed Oct. 10, 2019, which is a continuation of U.S. patent application Ser. No. 15/181,293, filed Jun. 13, 2016 and issued as U.S. Pat. No. 10,540,094, which is a continuation of U.S. patent application Ser. No. 14/732,507, filed Jun. 5, 2015 and issued as U.S. Pat. No. 9,367,486, which is a continuation of U.S. patent application Ser. No. 13/951,169, filed Jul. 25, 2013 and issued as U.S. Pat. No. 9,063,850, which is a continuation of U.S. patent application Ser. No. 13/645,588, filed Oct. 5, 2012 and issued as U.S. Pat. No. 8,601,228, which is a continuation of U.S. patent application Ser. No. 12/039,672, filed Feb. 28, 2008 and issued as U.S. Pat. No. 8,307,180, all of which are incorporated herein by reference.

FIELD OF INVENTION

The present invention relates generally to memory devices. In particular, the present invention relates to providing systems, methods and devices for run-time configuration of mass memory devices.

BACKGROUND OF THE INVENTION

In a typical environment involving digital data processing and/or data communications, memory devices are invoked for a variety of reasons, for example, to read, write, modify, delete, or change the attributes of the data that resides on a memory device. These operations (hereinafter referred to as memory 'access' operations) may be targeted to access varying chunks of data according to the needs of an application program that invokes the specific memory access operation. For example, an application may require access to a small chunk of data from random addresses, the same address, or sequential addresses on the memory device. Similarly, the same or a different application may require access to large chunks of data from random addresses, the same address, or sequential addresses on the memory device. Examples of the different applications that may access a memory device include file systems, different databases, kernel reading code pages, and other applications that use the memory device.

It is often the case that a mass memory device is optimized for one kind of application, or a defined group of applications, with particular memory access characteristics. This optimization, for example, may entail optimization of data throughput, life time and/or power consumption associated with the memory device. Due to this fixed optimization strategy, when a memory device is placed into a different environment with new access demands, it may fail to optimally perform under the requirements of the new environment. The lack of flexibility in optimizing such memory devices may be partly due to inherent limitations that render these memory devices incapable of accommodating optimized functionalities for multiple kinds of access

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operations. In other cases, however, the reason for electing to a memory device for a defined, and thus limited, group of applications is to simplify the design, and to effect cost savings. In addition, it is generally very difficult for a memory device to predict access requirements that are necessitated by yet-to-be determined future application needs.

SUMMARY OF THE INVENTION

A method, system and memory device are therefore provided to overcome the deficiencies of the prior art systems by allowing run-time configuration of a mass memory device. In one embodiment of the present invention a method for configuring access to a memory device is provided. The method comprises receiving one or more commands for activating one or more access profiles associated with the memory device, and configuring access to the memory device in accordance with at least one of the access profiles. The access profiles may correspond to at least one of a random and a sequential mode of access. The access profiles may further correspond to at least one of a read, a write, an erase, and a modify attribute operation.

In another embodiment of the present invention, one or more access profiles are adapted to accommodate repeated access requests to an identical address of the memory device. In another embodiment, one or more access profiles are adapted to produce an optimized performance associated with the memory device. Furthermore, the performance may be optimized in accordance with at least one of: data throughput, lifetime, and power consumption associated with the memory device.

In another embodiment of the present invention, one or more received commands comprise a metadata portion for designating a preferred access profile corresponding to the command. Furthermore, a specific memory location may be utilized in accordance with the access profile. In one embodiment, the specific memory location may comprise a section of the memory device with special characteristics. For example, it may include a more durable and performance-effective portion of the physical memory, or a portion of the memory that utilizes a specific memory technology. In another embodiment, the specific memory location may comprise a separate physical memory chip.

In another embodiment of the present invention, one or more access profiles are associated with one or more partitions of the memory device. Yet, in another embodiment, the configuring of the memory device is adapted in parallel for two or more parallel access profiles. In one embodiment, such configuring is carried out in accordance with JESD84 standard for eMMC. This configuring may further comprise designating access priority levels to resolve simultaneous access conflicts to memory resources. In another embodiment of the present invention, the memory device is used to effect both mass memory and system memory implementations. In another embodiment, a default access profile may be used to configure the memory device upon power up.

Another aspect of the present invention relates to a memory device that comprises one or more registers for storing one or more predefined access profiles associated with the memory device. The memory device also comprises receiving means for receiving one or more commands for activating one or more access profiles associated with the memory device, and configuring means for configuring access to the memory device in accordance with at least one of the predefined access profiles. In another embodiment, a currently active access profile may reside in a designated

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memory register. In another embodiment, one or more of the predefined access profiles may be updated with a new version of the access profile.

In another embodiment of the present invention, a computer program product embodied on a computer-readable medium is disclosed. The computer program product comprises a computer code for receiving one or more commands for activating one or more access profiles associated with the memory device, and a computer code for configuring access to the memory device in accordance with at least one of the access profiles. In another embodiment, a system for accessing a memory device is disclosed. The system comprises an entity for receiving one or more commands for activating one or more access types associated with the memory device, and an entity for configuring access to the memory device in accordance with at least one of the access profiles. In another embodiment, a system for accessing a memory device is disclosed. The system comprises a host for issuing one or more commands in accordance with access needs for the memory device, and an entity for receiving the commands and configuring access to the memory device in accordance with at least one or more access profiles.

Those skilled in the art will appreciate that various embodiments discussed above, or parts thereof, may be combined in a variety of ways to create further embodiments that are encompassed by the present invention.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a perspective view of an exemplary electronic device within which various embodiments of the present invention may be implemented;

FIG. 2 illustrates an exemplary schematic representation of the circuitry which may be included in the electronic device of FIG. 1.

FIG. 3 illustrates a flow diagram of an exemplary embodiment of the present invention.

FIG. 4 illustrates a flow diagram of another exemplary embodiment of the present invention.

FIG. 5 illustrates an exemplary device in accordance with an embodiment of the present invention.

DETAILED DESCRIPTION OF VARIOUS EMBODIMENTS

In the following description, for purposes of explanation and not limitation, details and descriptions are set forth in order to provide a thorough understanding of the present invention. However, it will be apparent to those skilled in the art that the present invention may be practiced in other embodiments that depart from these details and descriptions.

The problem of configuring a memory device for use in different environments has been traditionally addressed by using separate memory devices in systems with different use cases. For example, a system may utilize a mass memory device separate from a system memory device to accommodate different memory access demands.

The various embodiments of the present invention disclose methods, systems and devices to enable run-time configuration of a memory device in accordance with certain memory access profiles. The configuration may be effected for a portion of the memory device, a partition of the memory device, or even one single access location on the memory device. Since the system that accesses the memory device knows, or is capable of determining, the type of memory access needs (e.g., whether it is a read, write, erase, modify attribute, random, or a sequential operation), it can

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issue commands for configuring the memory device in accordance with an access profile that is most optimized/suitable for the particular access command. Such access profiles, for example, may be adapted for optimizing data throughput, lifetime and/or power consumption associated with particular uses of the memory device. In addition, according to the embodiments of the present invention, a default access profile may be defined to configure a memory device when, for example, the device or system initially boots up. Such a default profile, while providing a starting point for potential future modifications, may be pre-selected to accommodate the most likely access needs for that memory device. This profile may remain in effect until the memory device is powered down, or it may be replaced by another profile in accordance with the embodiments of the present invention.

In accordance with embodiments of the present invention, the information regarding the nature and type of memory access allows the memory device to organize itself in a manner that is most suited for a particular access command, resulting in improved performance and higher reliability. These improvements are largely due to the elimination of background operations and unnecessary data merging that are normally associated with traditional memory access methods. Although effective in both the random and sequential memory access modes, the techniques of the various embodiments of the present invention may be more effective in optimizing sequential memory access operations, where background processing and data merging are more abundant. These optimizations further extend the life of the storage device, and result in reduced energy consumption by the device.

The embodiments of the present invention further enable the utilization of the same memory device both as the mass storage memory and the system memory, thus eliminating the need for separate memory devices that are utilized in the systems of prior art. For example, all non-volatile memory needs of a system may be accommodated using a single eMMC memory, where the Operating System image, user data, and other parameters may be stored on the same device. Similarly, in multimedia applications that require very high density mass storage devices (e.g., in the order of several Gigabytes), the very same memory device may be used to store the various types of user applications, the Operating System and other system data files. This consolidation is expected to further spur the adoption of a standardized memory device with higher production volumes, and to eventually lead to lower-cost memory devices. The advent of such cost-effective, single-memory devices are particularly beneficial to the development of mobile devices in which size and cost constraints are most significant.

In accordance with one embodiment of the present invention, as illustrated in FIG. 5, a memory device 500 may comprise a physical memory 502 with one or more registers 504 for accommodating the predefined access profiles that are used to optimize the memory device. The memory device 500 may further comprise a receiving means 510 that is adapted to receive one or more commands, through the communication interface 512, for activating a particular access profile. To facilitate understanding of the present embodiment, the receiving means 510 is illustrated as comprising a separate section of the controller 508. However, it is understood that the receiving means 510 and the controller 508 may also be implemented as a single entity. Upon the receipt of one or more commands, the controller 508 may configure the memory device 500 in accordance with one or more access profiles that reside in memory registers 504.

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The communication between the controller **508** and the physical memory **502** may be conducted through the interface **506**.

By the way of example, and not limitation, one predefined access profile may be a burst mode profile that facilitates high-speed transfer of large data chunks and provides a 'ready' indication to the host prior to, or after, such transfer. In order to minimize the transfer time, the needed flash memory management operations may take place subsequent to the transfer at a convenient time, for example, while no other activities or memory access operations are taking place. Another example of an access profile includes a random mode profile which enables quick access to short, random memory locations on the device. The memory device in accordance with embodiments of the present invention may further comprise another register for accommodating the currently active access profile. This profile, which may be any one of the supported predefined profiles, governs the current access operations to the memory device. For example, such register may comprise a default profile that is activated during the boot up of the host system and/or the power up of the memory device. This active profile may remain in effect until the memory device is powered down, or it may be replaced by another profile in accordance with the embodiments of the present invention. Run-time configurability of the memory device in accordance with the present invention is effected by replacing the contents of the currently active profile register with one of the predefined profiles that resides on the first set of registers. Accordingly, when the need for a new type of memory access arises, a command may be issued to activate a suitable profile. The command may activate any one of the predefined access profiles, including but not limited to, the default profile.

In accordance with another embodiment, the various access profiles may be updated or uploaded onto the memory device. For example, an existing access profile may be augmented (or completely replaced with a new version) to add or remove certain features and functionalities. Alternatively, or additionally, an entirely new access profile may be uploaded to the memory device, thus increasing the number of available access profiles that can be readily used to configure the memory device. By the way of example, and not by limitation, an access profile may be implemented as a binary file that further comprises the required logic to implement an access profile. This way, the access profile may be considered part of the memory device firmware responsible for handling specific accesses needs in an optimized fashion.

FIGS. 1 and 2 show one representative electronic device **12** within which embodiments of the present invention may be implemented. It should be understood, however, that the present invention is not intended to be limited to one particular type of device. In fact, the various embodiments of the present invention may be readily adapted for use in any stand-alone or embedded system that comprises or accesses a memory device. The electronic device **12** of FIGS. 1 and 2 includes a housing **30**, a display **32** in the form of a liquid crystal display, a keypad **34**, a microphone **36**, an ear-piece **38**, a battery **40**, an infrared port **42**, an antenna **44**, a smart card **46** in the form of a UICC according to one embodiment, a card reader **48**, radio interface circuitry **52**, codec circuitry **54**, a controller **56** and a memory **58**. Individual circuits and elements are all of a type well known in the art, for example in the Nokia range of mobile telephones.

FIG. 3 is an example flow diagram illustrating run-time configurability of a memory device in accordance to an

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embodiment of the present invention. As illustrated in FIG. 3, upon boot up of the system in step **100**, the memory device in accordance with embodiments of the present invention organizes itself according to the default profile in step **102**. The exemplary default profile used in FIG. 3 configures the memory device to accommodate the reading of large sequential data from the memory device. In step **104**, the system reads a large amount of sequential data, which for example, may comprise the operating system of the host device. Upon completion of the large read operation, the system enters an idle state in step **106**. Since the majority of memory access operations during an idle state is likely to involve short random read/write operations, the memory device, in step **108**, is commanded to activate an access profile for reading/writing short random data. In Step **110**, the system requires large sequential reads/writes. By the way of example, and not by limitation, this need may arise when the system is connected to an external mass storage device. Such a mass storage device may, for example, include a stand-alone memory device such as a USB memory, or a PC or other electronic device that comprises one or more mass storage components. In anticipation of large data transfers to/from the external memory device, the memory device in accordance with embodiments of the present invention, in step **112**, receives a command to activate the access profile that is optimized for reading/writing large sequential data. In step **114**, the system conducts at least a portion of the large sequential read/write transfer. While the large data access operations may be completed without further interruptions, in one exemplary embodiment, the system of the present invention may need to access the memory device in short, random I/O access cycles, as illustrated in step **116**. In accordance with one embodiment of the present invention, in step **118**, the memory device may receive a command to suspend its current access profile, which is directed towards reading/writing long sequential data, and activate an alternate access profile that is optimized for reading/writing short random data. Once the system completes short memory access operations in step **120**, the memory device, in step **122**, may receive a subsequent command to revert back to the access profile for reading/writing large sequential data. The system may then resume reading/writing large sequential data in step **124**.

As describe above, the example embodiment of the present invention as illustrated in FIG. 3 suspends the large data transfer while conducting short I/O access operations. However, in some applications, it may be advantageous to conduct two or more memory access operations in parallel. To this end, FIG. 4, illustrates an alternate embodiment of the present invention according to which two or more memory access operations (and their corresponding access profiles) may be implemented in parallel. In FIG. 4, steps **200** to **216** represent similar operations as their counterparts in FIG. 3. Specifically, upon boot up in step **200**, the memory device in accordance with embodiments of the present invention organizes itself according to the default profile in step **202**. The exemplary default profile used in FIG. 4 configures the memory device to accommodate the reading of large sequential data from the memory device. In step **204**, the system reads a large amount of sequential data, which for example, may comprise the operating system of the host device. Upon completion of the large read operation, the system enters an idle state in step **206**. Since the majority of memory access operations during an idle state is likely to involve short random read/write operations, the memory device, in step **208**, is commanded to activate an

access profile for reading/writing short random data. The system may then require access to large sequential reads/writes in step 210. This need may arise, for example, in preparation for large data transfers to/from an external memory device. The memory device in accordance with embodiments of the present invention, in step 212, receives a command to activate the access profile that is optimized for reading/writing large sequential data. In step 214, the system conducts at least a portion of the large sequential read/write transfers before the system need for short read/write access cycles to the memory device arises in step 216. In contrast to the example embodiment of the present invention in accordance with FIG. 3, the present embodiment in accordance with FIG. 4 accommodates both memory access modes by commanding the memory device in accordance with embodiments of the present invention to activate a parallel access profile for reading/writing short random data in step 220. Accordingly, while the system continues to read/write large sequential data in step 218, it may simultaneously (or in an interleaved fashion) conduct short memory access operations in step 222.

While the embodiment of the present invention in accordance with FIG. 4 was described in terms of only two simultaneous access profiles, it is understood that similar operations may be carried out to allow the implementation of more than two access profiles in parallel. One specific parallel implementation of memory access profiles may be realized in a format that is compatible with the current JEDEC JC64 eMMC version 4.3 (JESD84). JEDEC eMMC is a standardized mass storage device comprising a memory and a controller device. The controller handles block-management functions associated with the memory such as logical block allocation and wear leveling. The communication between the memory and the host device is also handled by the controller according to a standard protocol. This protocol defines, among other signals, a bidirectional command signal, CMD, that is used for device initialization, and transfer of commands between the host and memory device. More specifically, CMD23 (SET_BLOCK_COUNT) defines the number of blocks (read/write) and the reliable writer parameter (write) for a block read/write command. CMD23 includes a 32 bit argument field, of which bits 15 to 0 are allocated for setting the number of blocks for the corresponding read/write command, and bits 30 to 16 are designated as stuff bits. In accordance to one embodiment of the present invention, these stuff bits may be utilized to designate different access profiles for the memory device. By the way of example, and not by limitation, one profile may be defined as a burst profile mode, corresponding to a fast, contiguous data access mode.

When in burst profile mode, the memory device, immediately after receiving all the data, may indicate "exit busy" and set the transfer mode to "transfer state," thus facilitating faster execution of subsequent accesses by the host. In addition, while the commands corresponding to the first access profile are still being executed, the memory device may also enable the host to send additional commands corresponding to a different access profile. This way, a degree of parallelism in the I/O operations is established. Furthermore, access priority levels may be defined to resolve access conflicts, where two or more profiles run in parallel and require access to the same memory resource at the same time. Examples of such a memory resources include a RAM buffer, a Flash bus, and other memory resources.

In accordance with another embodiment of the present invention, the access profile associated with a media device

may be adapted to comprise different control and/or setting profiles that are associated with different partitions of the memory device. Such partitions may comprise logical or physical partitions of the memory device. For example, one partition may be configured for random read/write operations while another partition may be configured to provide sequential access.

In accordance with another embodiment of the present invention, a memory access (e.g., an I/O read/write) command may be configured to comprise a metadata portion for designating a preferred access profile corresponding to that access command. For example, the system in accordance with the present invention may recognize that one address is being continuously and frequently updated, and accordingly, it may set an appropriate access profile for that memory command. The memory device—depending on its internal implementations and capabilities—may map such sustained and specific access operations to certain sections of the physical memory with special characteristics. For example, the mapping may be directed to a more durable and performance-effective portion of the physical memory, a portion of the memory that utilizes a specific memory technology, or to a separate physical chip that is more suitably designed for such repeated access operations. Thus, the memory device firmware may take an action in accordance with the access profile request of an embodiment of the present invention and handle the I/O operation in a different way.

The various embodiments of the present invention are equally applicable to both the embedded memory devices, such as NAND, mass memory, XiP, and similar devices, as well as to removable memory cards.

The various embodiments described herein are described in the general context of method steps or processes, which may be implemented in one embodiment by a computer program product, embodied in a computer-readable medium, including computer-executable instructions, such as program code, executed by computers in networked environments. A computer-readable medium may include removable and non-removable storage devices including, but not limited to, Read Only Memory (ROM), Random Access Memory (RAM), compact discs (CDs), digital versatile discs (DVD), etc. Generally, program modules may include routines, programs, objects, components, data structures, etc. that perform particular tasks or implement particular abstract data types. Computer-executable instructions, associated data structures, and program modules represent examples of program code for executing steps of the methods disclosed herein. The particular sequence of such executable instructions or associated data structures represents examples of corresponding acts for implementing the functions described in such steps or processes.

The foregoing description of embodiments has been presented for purposes of illustration and description. The foregoing description is not intended to be exhaustive or to limit embodiments of the present invention to the precise form disclosed, and modifications and variations are possible in light of the above teachings or may be acquired from practice of various embodiments. The embodiments discussed herein were chosen and described in order to explain the principles and the nature of various embodiments and its practical application to enable one skilled in the art to utilize the present invention in various embodiments and with various modifications as are suited to the particular use contemplated. The features of the embodiments described

herein may be combined in all possible combinations of methods, apparatus, modules, systems, and computer program products.

What is claimed is:

1. A memory device comprising:
 - a memory configured to be accessed according to a plurality of predefined access types, the plurality of predefined access types including at least a default access type; and
 - a controller configured to:
 - configure access to the memory device according to the default access type such that at least a portion of the memory is configured according to the default access type for at least one usage;
 - receive, from a host device, a first command to configure access to the memory device in accordance with a second access type of the plurality of predefined access types such that the at least the portion of the memory is configured according to the second access type;
 - configure, based at least in part on the first command, access to the at least the portion of the memory according to the second access type; and
 - receive, after the first command, at least one second command to write data to the at least the portion of the memory according to the second access type.
2. The memory device of claim 1, wherein the controller is further configured to:
 - configure access to the default access type during at least one of boot-up of the host device or power-up of the memory device.
3. The memory device of claim 1, wherein the default access type corresponds to a most likely access need of the memory device.
4. The memory device of claim 1, wherein the default access type is optimized for at least one of: data throughput, data lifetime, or power consumption associated with storing data.
5. The memory device of claim 1, wherein the default access type is preselected.
6. The memory device of claim 1, wherein the default access type includes at least one of: a sequential read, a sequential write, a random read, or a random write.
7. The memory device of claim 1, wherein the controller is further configured to:
 - activate, based at least in part on the first command, the second access type.
8. The memory device of claim 1, wherein the at least the portion of the memory includes one or more blocks and the default access type is optimized for data lifetime.
9. The memory device of claim 1, further comprising:
 - a register that includes the default access type.
10. The memory device of claim 1, wherein the controller is further configured to:
 - receive, from a host device, a third command to configure access to the memory device in accordance with a third access type of the plurality of predefined access types such that the at least the portion of the memory is configured according to the third access type; and
 - configure, based at least in part on the third command, access to the at least the portion of the memory according to the third access type.

11. A method comprising:
 - configuring, by a controller of a memory device with a memory, access to the memory device according to a default access type such that at least a portion of the memory is configured according to the default access type for at least one usage, wherein the default access type is one of a plurality of predefined access types;
 - receiving, from a host device and by the controller, a first command to configure access to the memory device in accordance with a second access type of the plurality of predefined access types such that the at least the portion of the memory is configured according to the second access type;
 - configuring, by the controller and based at least in part on the first command, access to the at least the portion of the memory according to the second access type; and
 - receiving, by the controller and after the first command, at least one second command to write data to the at least the portion of the memory according to the second access type.
12. The method of claim 11, further comprising:
 - configuring, by the controller, access to the default access type during at least one of boot-up of the host device or power-up of the memory device.
13. The method of claim 11, wherein the default access type corresponds to a most likely access need of the memory device.
14. The method of claim 11, wherein the default access type is optimized for at least one of: data throughput, data lifetime, or power consumption associated with storing data.
15. The method of claim 11, further comprising:
 - activating, based at least in part on the first command, the second access type.
16. The method of claim 11, wherein the memory device includes a register that includes the default access type.
17. The method of claim 11, wherein the at least the portion of the memory includes one or more blocks and the second access type is optimized for data lifetime.
18. The method of claim 11, wherein the default access type is preselected.
19. A host device, comprising:
 - an interface to couple the host device to a memory device having a memory configured to be accessed according to a plurality of predefined access types, the plurality of predefined access types including a default access type and a second access type; and
 - a controller configured with logic instructions to:
 - send first data to the memory device, wherein the first data is stored in a portion of memory according to the default access type;
 - send at least one command to configure access to the portion of the memory in accordance with the second access type for at least one usage, wherein the second access type replaces the default access type, wherein the default access type is preselected; and
 - send second data to the memory device, wherein the second data is stored according to the second access type.
20. The host device of claim 19, wherein the controller is further configured with the logic instructions to:
 - read one or more registers of the memory device to determine an indication of the plurality of predefined access types.